

A Simplified MMD Variable-Selection Procedure

Regularized Test-Power Optimization and Path Aggregation

1 Goal

Let P and Q be two distributions on $\mathbb{R}^d$. The goal is to identify a sparse set of variables whose joint distribution captures the discrepancy between P and Q .

Let $a = (a_1, \dots, a_d) \in \mathbb{R}_+^d$ denote nonnegative ARD weights in a product kernel k_a . Let $\hat{J}_n(a)$ denote the empirical MMD test-power criterion used to tune the kernel. Larger values of $\hat{J}_n(a)$ correspond to larger estimated test power.

The proposed simplification has only two essential ingredients:

$$\boxed{\text{regularized test-power optimization}} \quad + \quad \boxed{\text{aggregation over the regularization path}}.$$

No validation-power weighting or auxiliary permutation test is required in the basic procedure.

2 Regularized optimization along a path

Choose a finite grid of regularization parameters

$$\Lambda = \{\lambda_1, \dots, \lambda_L\}, \quad 0 < \lambda_1 < \dots < \lambda_L.$$

For each $\lambda \in \Lambda$, compute

$$\hat{a}_\lambda \in \arg \max_{a \in \mathbb{R}_+^d} \left\{ \log \hat{J}_n(a) - \lambda \|a\|_1 \right\}. \quad (1)$$

The ℓ_1 penalty suppresses variables that do not improve test power, while small values of λ allow weaker discriminating variables to enter.

It is not necessary to choose a single “best” value of λ . Instead, all solutions along the path are retained.

3 Path aggregation

For each variable j , define an aggregated importance score

$$\hat{\Pi}_j = \sum_{\lambda \in \Lambda} w_\lambda \rho(\hat{a}_{\lambda,j}), \quad (2)$$

where $w_\lambda \geq 0$ and $\sum_{\lambda \in \Lambda} w_\lambda = 1$. The simplest choice is uniform weighting,

$$w_\lambda = \frac{1}{|\Lambda|}.$$

The transformation ρ should satisfy

$$\rho(0) = 0, \quad \rho(t) > 0 \quad \text{for } t > 0.$$

A particularly simple choice is

$$\rho(t) = t.$$

A bounded alternative is $\rho(t) = t/(1+t)$.

The key idea is that a true discriminating variable need not be selected for every value of λ . It is sufficient that it becomes important somewhere along the regularization path. Aggregation therefore avoids committing to one tuning parameter and naturally accommodates variables of different signal strengths.

4 Final variable selection

Given a threshold $\tau > 0$, define

$$\widehat{S} = \left\{ j \in \{1, \dots, d\} : \widehat{\Pi}_j > \tau \right\}. \quad (3)$$

Thus the basic procedure has only two statistical tuning components:

- (i) the regularization path Λ ;
- (ii) the final aggregation threshold τ .

In practice, Λ can be a modest log-spaced grid spanning sparse to weakly regularized solutions. The threshold τ controls how much finite-sample noise in the aggregated weights is tolerated.

5 Optional stability enhancement

Data splitting is *not* required by the basic method. Equation (1) can be optimized using the full sample for every λ .

If additional finite-sample stability is desired, repeated subsampling may be added as an optional layer. For splits $b = 1, \dots, B$, compute $\widehat{a}_\lambda^{(b)}$ and aggregate

$$\widehat{\Pi}_j = \frac{1}{B} \sum_{b=1}^B \sum_{\lambda \in \Lambda} w_\lambda \rho\left(\widehat{a}_{\lambda,j}^{(b)}\right). \quad (4)$$

This should be viewed as a stability device rather than as an essential cross-validation step. The default formulation remains the full-data path aggregation in Eq. (2).

6 Connection with the population theory

The simplified method mirrors the population argument directly. Let

$$a_\lambda^\star \in \arg \max_{a \in \mathbb{R}_+^d} \{\log J(a) - \lambda \|a\|_1\}$$

and define

$$\Pi_j = \sum_{\lambda \in \Lambda} w_\lambda \rho(a_{\lambda,j}^\star).$$

The intended theory has two components:

1. **Irrelevant-variable exclusion:** if $j \notin S^\star$, then $a_{\lambda,j}^\star = 0$ for every λ .
2. **Path coverage:** if $j \in S^\star$, then $a_{\lambda,j}^\star > 0$ for at least one $\lambda \in \Lambda$ with positive weight.

These immediately imply

$$\boxed{\Pi_j > 0 \iff j \in S^\star.}$$

Hence path aggregation is not an additional heuristic: it is the empirical counterpart of the population support-recovery argument.

7 Simplification relative to the current MMD-CV procedure

The proposed basic method removes several auxiliary components:

Current component	Simplified treatment
Training/validation split	Not required by default
Validation-power weighting	Removed
Auxiliary permutation test / p -value filter	Removed
Selection of a single λ	Replaced by path aggregation
Multiple aggregation heuristics	Single score $\hat{\Pi}_j$

The resulting method can therefore be summarized in four steps.

Algorithm 1 Simplified MMD variable selection

Require: Samples from P and Q , regularization grid Λ , threshold τ .

- 1: **for** $\lambda \in \Lambda$ **do**
 - 2: Compute $\hat{a}_\lambda$ by solving Eq. (1).
 - 3: **end for**
 - 4: Compute $\hat{\Pi}_j$ for $j = 1, \dots, d$ using Eq. (2).
 - 5: Return $\hat{S} = \{j : \hat{\Pi}_j > \tau\}$.
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Implementation note. A proximal-gradient or related sparse optimizer may be used to obtain exact zeros in $\hat{a}_\lambda$, but this is an implementation choice rather than part of the statistical definition of the method.